

Shift spaces on groups: computability and dynamics

Calculabilité et dynamique des sous-décalages sur des groupes

Sebastián Andrés **Barbieri Lemp**

LIP, ENS de Lyon

Soutenance de Thèse
28 Juin 2017

Motivation

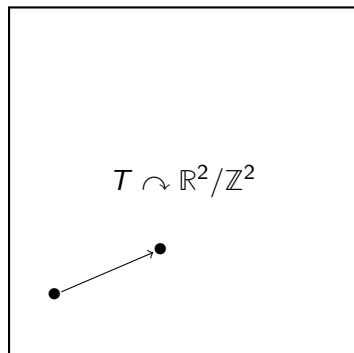
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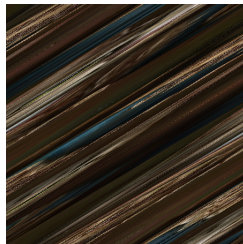
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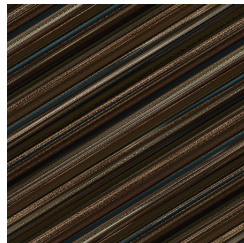
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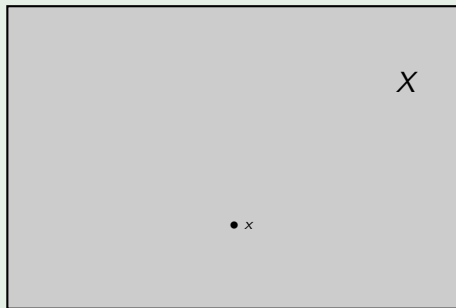
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A dynamical system might be complicated. A good idea is to code its trajectories using a partition.

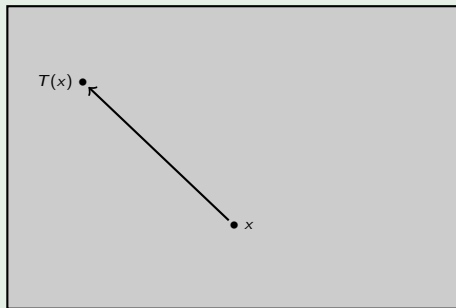
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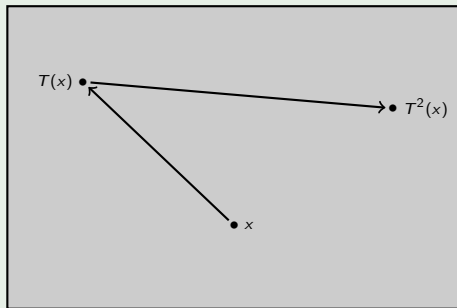
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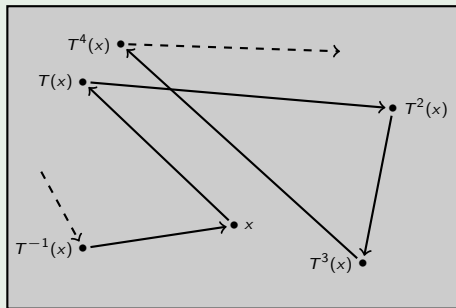
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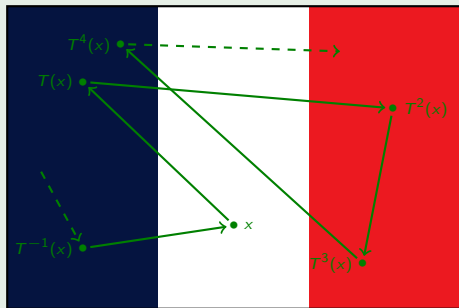
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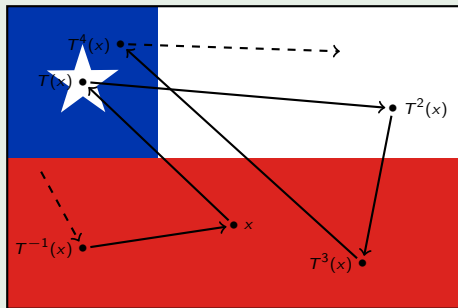


$$\varphi(x) = \cdots \begin{array}{c} \text{dark blue} \quad \text{white} \quad \text{dark blue} \quad \text{red} \quad \text{red} \quad \text{dark blue} \\ -1 \quad 0 \quad 1 \quad 2 \quad 3 \quad 4 \end{array} \cdots \in \{\text{red}, \text{dark blue}, \text{white}\}^{\mathbb{Z}}$$

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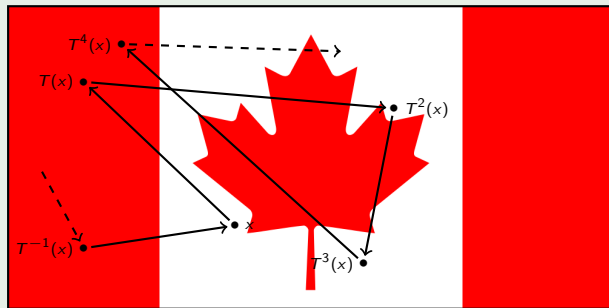
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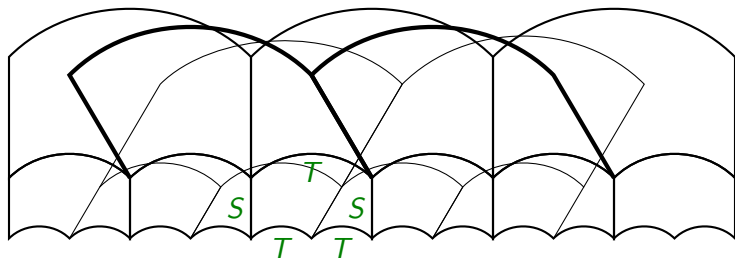
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$$2 \times (x + 1) = (2 \times x) + 2 \quad \rightsquigarrow \quad ST = T^2S$$

Why a group action?

$$\langle S, T \mid ST = T^2S \rangle \cong BS(1, 2)$$



- ▶ G is a finitely generated group.
- ▶ $\mathcal{A}$ is a finite alphabet. Ex: $\mathcal{A} = \{0, 1\}$.
- ▶ $\mathcal{A}^G$ is the set of configurations, $x : G \rightarrow \mathcal{A}$
- ▶ $\sigma : G \times \mathcal{A}^G \rightarrow \mathcal{A}^G$ is the left shift action given by:

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Definition: full G -shift

The pair $(\mathcal{A}^G, \sigma)$ is called the *full G -shift*.

Example

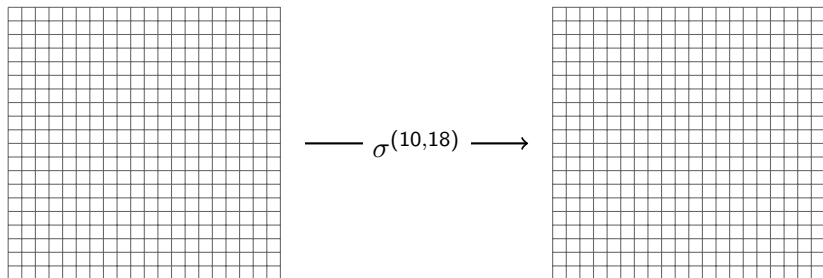


Figure: A random configuration $x \in \{\blacksquare, \square\}^{\mathbb{Z}^2/20\mathbb{Z}^2}$ and its image by $\sigma^{(10,18)}$.

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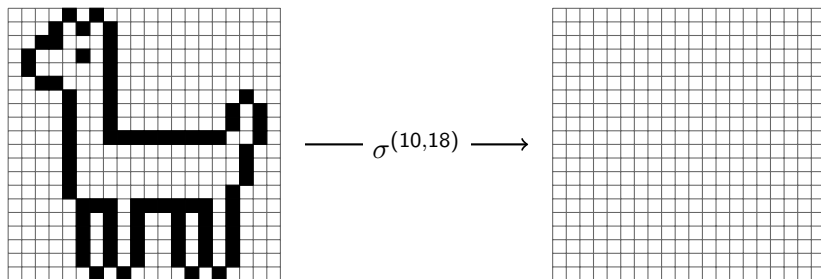


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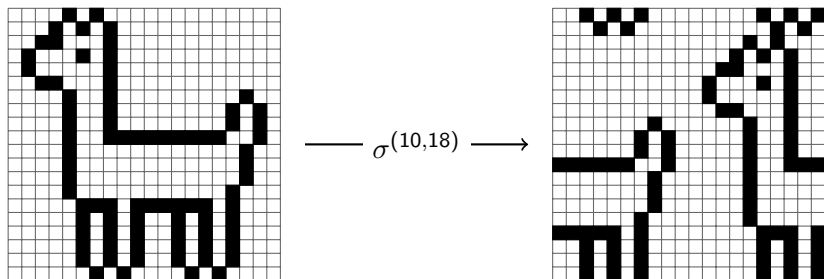


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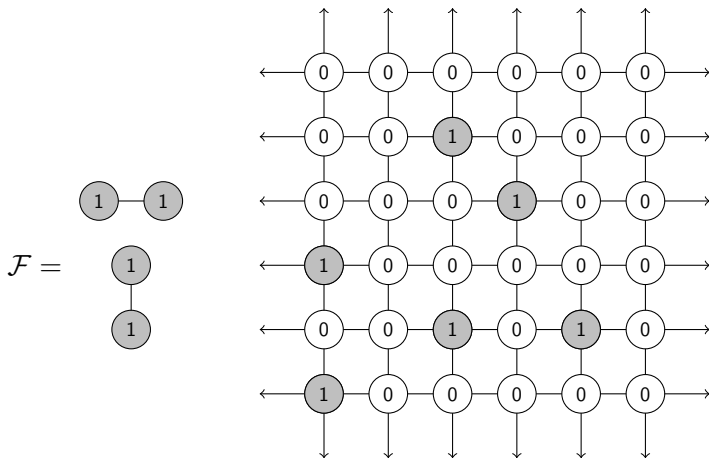
Proposition

A subshift is a set of configurations avoiding patterns from a list $\mathcal{F}$.

$$p \in \mathcal{A}^S, \quad [p] = \{x \in \mathcal{A}^G \mid x|_S = p\}$$
$$X = X_{\mathcal{F}} = \mathcal{A}^G \setminus \bigcup_{g \in G, p \in \mathcal{F}} \sigma^g([p])$$

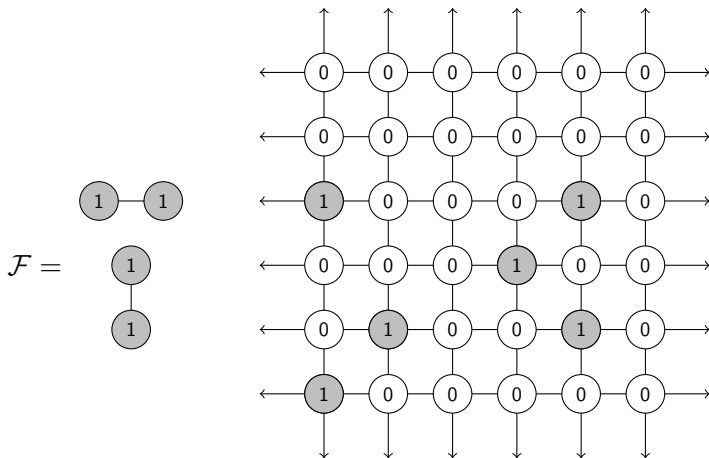
Example in $\mathbb{Z}^2$: Hard square shift

The set of assignments of $\mathbb{Z}^2$ to $\{0, 1\}$ such that there are no two adjacent ones.



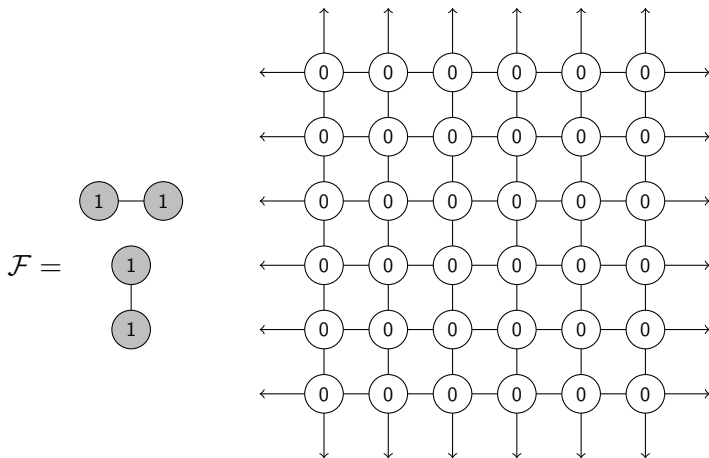
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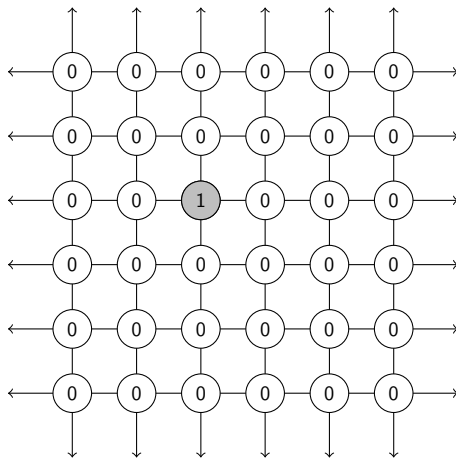
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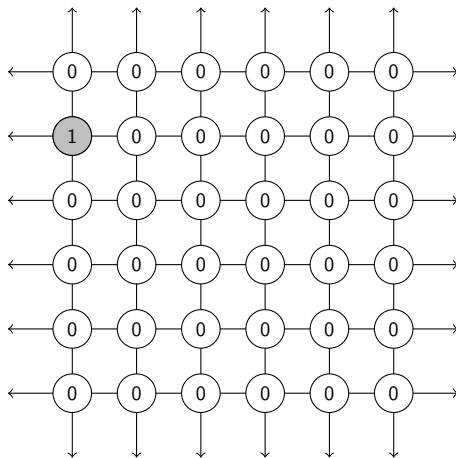
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$$X_{\leq 1} := \{x \in \{0,1\}^G \mid 0 \notin \{x_u, x_v\} \implies u = v\}.$$



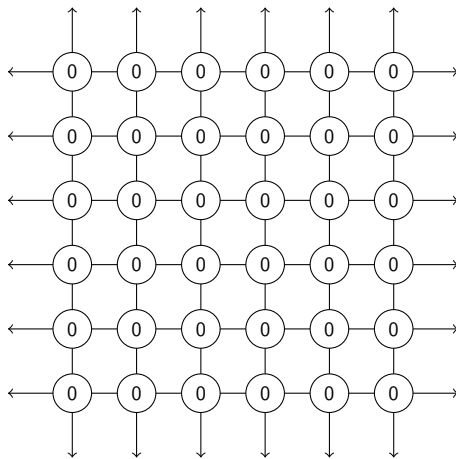
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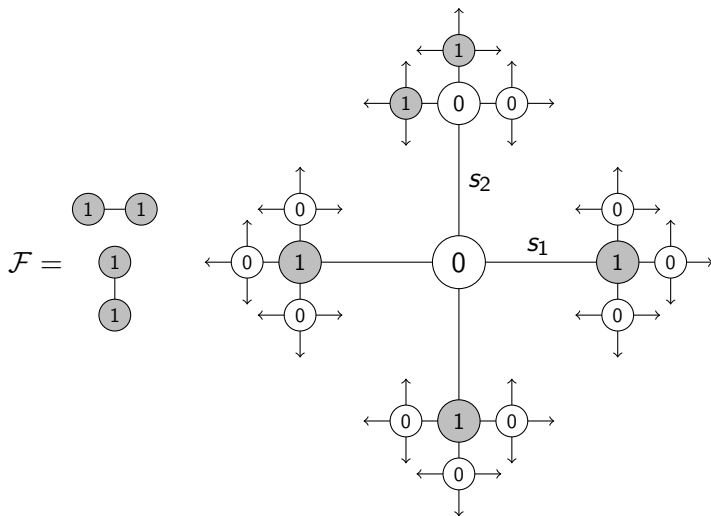


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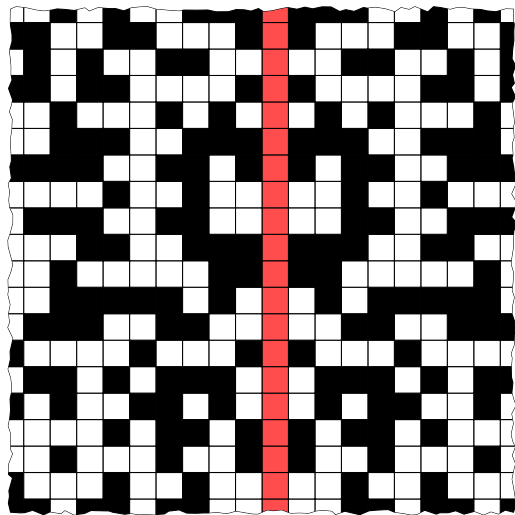
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Example: Hard square in F_2 .



Example: Mirror shift on $\mathbb{Z}^2$



Classes of subshifts

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- an *effectively closed subshift* if X can be defined by a recursively enumerable coding of a set of forbidden patterns.
Ex: Mirror shift in $\mathbb{Z}^2$.

Why computability and dynamics?

Dynamical objects

▷ Expansivity, periodic points, aperiodicity, invariant measures, entropy, recurrence, mixing, minimality, etc.

Computational objects

▷ Finite information, description by Turing machines, computability of invariants, domino problem, simulation, etc.

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The purpose of my thesis is to study the **interplay between computability and dynamics in shift spaces in groups**.

Some topics I worked on:

- Aperiodic subshifts (and SFTs) on groups.
- Realization of subshifts with uniform densities.
- Effectively closed subshifts, subactions and simulation theorems.
- Notions of computability for subshifts on groups.
- Computability in automorphism groups and the topological full group of a shift space.

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Subactions, effectiveness, simulation and aperiodicity.

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There is $C > 0, x \neq y \implies \exists g \in G, d(T^g(x), T^g(y)) > C$

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Shift spaces

Every shift space is expansive

If $x \neq y$, there is $g \in G$ such that $x_g \neq y_g$. We have

$$\sigma^{g^{-1}}(x)|_{1_G} \neq \sigma^{g^{-1}}(y)|_{1_G}.$$

What about the subactions of these classes?

Let $X \subset \mathcal{A}^G$ be a subshift and $H \leq_{\text{f.g.}} G$.

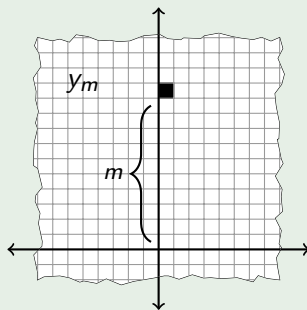
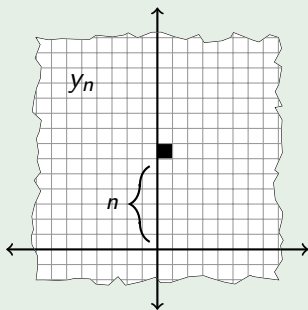
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- ▷ What can we say about the system $(X, \sigma|_H)$?
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Remark: Subshifts are expansive, subactions not necessarily



$$\inf_{n \neq m} d(\text{Orb}_{\sigma|_{\mathbb{Z}}}(y_n), \text{Orb}_{\sigma|_{\mathbb{Z}}}(y_m)) = 0$$

Question 1: which are the subactions?

Effectively closed action

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Every subaction of an effectively closed subshift (also sofic/SFT) is effectively closed.

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Before answering that question, let us motivate this kind of results:

The philosophy behind it

Finitely presented group

A group G is finitely presented if it can be described as $G = \langle S | R \rangle$ where both S and $R \subset (S \cup S^{-1})^*$ are finite.

$$\mathbb{Z}^2 \cong \langle a, b \mid aba^{-1}b^{-1} \rangle.$$

$$BS(1, 2) \cong \langle a, b \mid aba^{-2}b^{-1} \rangle.$$

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Recursively presented group

A group G is recursively presented if it can be described as $G = \langle S | R \rangle$ where $S \subset \mathbb{N}$ and $R \subset (S \cup S^{-1})^*$ are recursive sets.

$$L \cong \langle a, t \mid (at^n at^{-n})^2, n \in \mathbb{N} \rangle.$$

$$\bigoplus_{i \in \mathbb{N}} (\mathbb{Z}/2\mathbb{Z}) \cong \langle \{a_n\}_{n \in \mathbb{N}} \mid \{a_n^2\}_{n \in \mathbb{N}} \cup \{a_j a_k a_j^{-1} a_k^{-1}\}_{j, k \in \mathbb{N}} \rangle.$$

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Theorem (Highman 1961)

For every recursively presented group H there exists a finitely presented group G such that H embeds into G .

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“A complicated object is realized inside another object which admits a much simpler presentation.”

The philosophy behind it

Theorem (Highman 1961)

For every recursively presented group H there exists a finitely presented group G such that H embeds into G .

“A complicated object is realized inside another object which admits a much simpler presentation.”

word problem: is it algorithmically decidable if a word over a set of generators of a group represents the identity?

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There are finitely presented groups with undecidable word problem

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There are finitely presented groups with undecidable word problem

Just apply Highman's theorem to

$G = \langle a, b, c, d \mid b^{-n}ab^n = c^{-n}dc^n, n \in \text{HALT} \rangle \dots$ done!

Two recent simulation results

Theorem (Hochman 2009)

For every effectively closed action $T : \mathbb{Z}^d \curvearrowright X \subset \{0,1\}^{\mathbb{N}}$ there exists a $\mathbb{Z}^{d+2}$ -SFT $\hat{X}$ such that one of its $\mathbb{Z}^d$ -subactions is an extension of T .

$$\begin{array}{ccc} \mathbb{Z}^{d+2} & (\hat{X}, \sigma) & \\ & \downarrow \text{subaction} & \\ \mathbb{Z}^d & (\hat{X}, \sigma|_{\mathbb{Z}^d}) & \xrightarrow{\text{factor}} (X, T) \end{array}$$

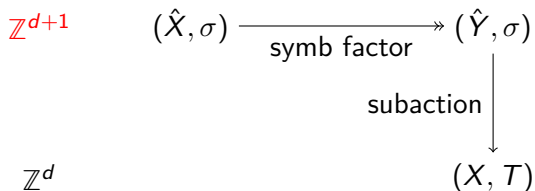
Moreover, the factor is small.

Two recent simulation results

One can do better in the expansive case:

Theorem (Aubrun-Sablik 2010, Durand-Romaschenko-Shen 2010)

Every effectively closed $\mathbb{Z}^d$ -subshift is the subaction (projective subaction) of a $\mathbb{Z}^{d+1}$ -sofic subshift.



An application: strongly aperiodic subshifts

Definition (Strongly aperiodic subshift)

A subshift $X \subset \mathcal{A}^G$ is *strongly aperiodic* if the shift action is free

$$\forall x \in X, \forall g \in G, \sigma^g(x) = x \Rightarrow g = 1_G.$$

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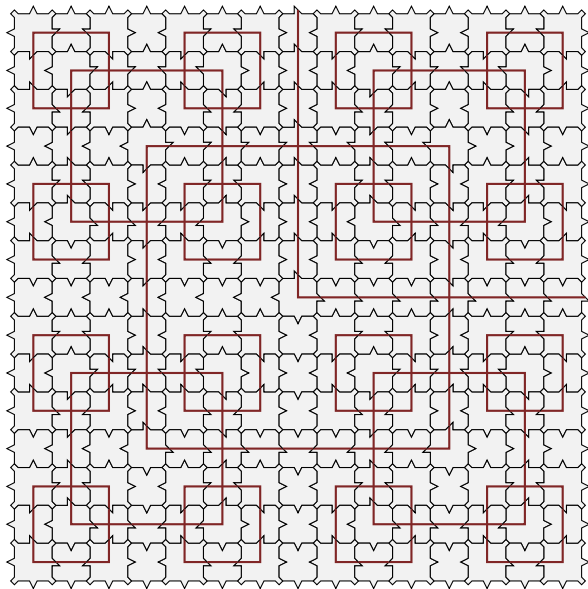
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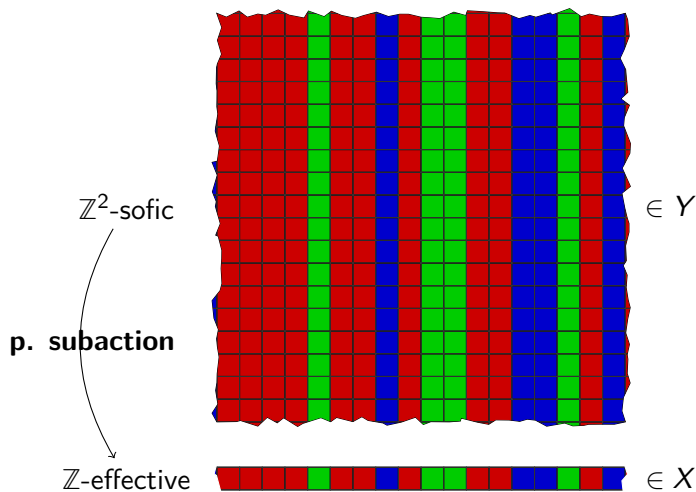
Theorem (Berger 1966, Robinson 1971, Kari 1996, Jeandel-Rao 2015)

There exist strongly aperiodic SFTs on $\mathbb{Z}^2$.

Example of strongly aperiodic $\mathbb{Z}^2$ -SFT: Robinson tileset



Simulation in the case of subshifts



So... why is simulation important?

It is complicated to come up with $\mathbb{Z}^2$ -SFTs which are strongly aperiodic, however, finding a $\mathbb{Z}$ -effectively closed subshift which is aperiodic is easy.

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Example

Let x be a fixed point of the Thue-Morse substitution.

$$0 \rightarrow 01 \rightarrow 0110 \rightarrow 01101001 \rightarrow 0110100110010110 \rightarrow \dots$$

Then $X = \overline{\text{Orb}_\sigma(x)}$ is strongly aperiodic and effectively closed.

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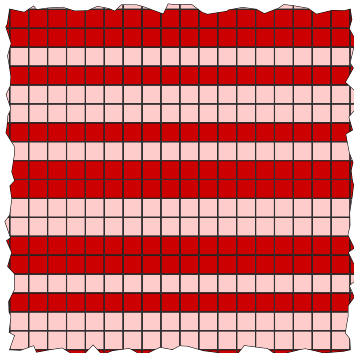
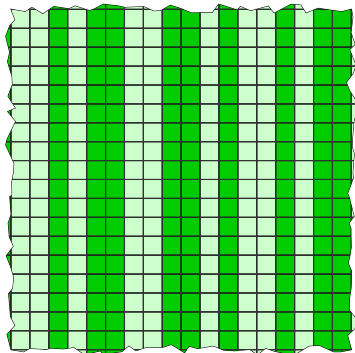
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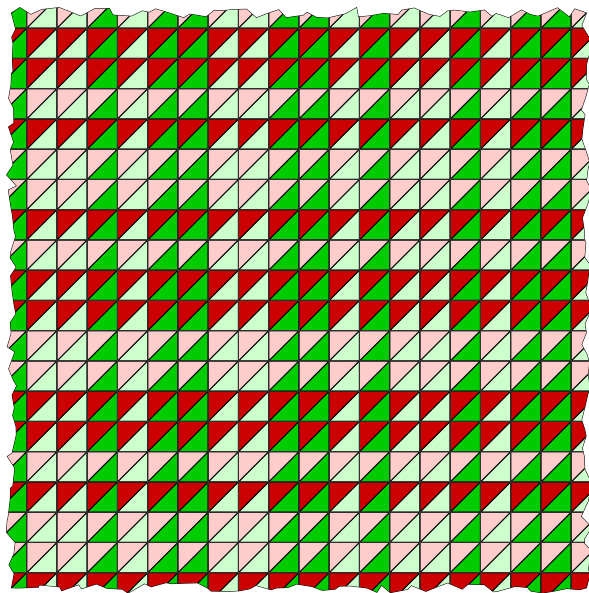
Example

A Sturmian subshift defined by an irrational computable slope α .

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Examples

- ▶ Easy construction of strongly aperiodic $\mathbb{Z}^2$ -SFTs

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- ▶ $\mathbb{Z}^2$ -SFTs with no computable configurations (Original result by Hanf-Myers 1974)

So... why is simulation important?

Examples

- ▶ Easy construction of strongly aperiodic $\mathbb{Z}^2$ -SFTs
- ▶ $\mathbb{Z}^2$ -SFTs with no computable configurations (Original result by Hanf-Myers 1974)
- ▶ Classifying the entropies of $\mathbb{Z}^2$ -SFTs (Original result by Hochman-Meyerovitch 2010)

What if we were able to do this in general groups?

Two new results in general groups

Let $T : G \curvearrowright X \subset \{0,1\}^{\mathbb{N}}$ be an effectively closed action of a finitely generated group.

Theorem (B-Sablik, 2016)

For any semidirect product $\mathbb{Z}^2 \rtimes G$ there exists a $\mathbb{Z}^2 \rtimes G$ -SFT such that its G -subaction is an extension of T .

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Theorem (B, 2017)

For any pair of infinite and finitely generated groups H_1, H_2 there exists a $(G \times H_1 \times H_2)$ -SFT such that its G -subaction is an extension of T .

Strongly aperiodic SFTs in these groups.

We want to go ahead and use the simulation theorems to produce strongly aperiodic SFTs, but...

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In fact...

Question by Glasner and Uspenskij 2009

- Is there a countable group which admits no aperiodic subshifts over the alphabet $\mathcal{A} = \{0, 1\}$?

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Theorem (Gao-Jackson-Seward, 2009)

No! All do.

And the proof is a little bit technical.

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Theorem (Gao-Jackson-Seward, 2009)

No! All do.

And the proof is a little bit technical.

Theorem (Aubrun-B-Thomassé, 2015)

No! All do.

But the proof is much shorter, moreover, the construction is effective for groups with decidable word problem. It is based on a probabilistic argument using Lovász local lemma.

Strongly aperiodic subshifts?

Moreover, a recent result by Jeandel states:

Theorem (Jeandel, 2015)

Let G be a recursively presented group. If G admits an effectively closed strongly aperiodic subshift then its word problem is decidable.

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With our result we can write:

Theorem (Aubrun-B-Thomassé, 2015)

Let G be a recursively presented group. G admits an effectively closed strongly aperiodic subshift if and only if its word problem is decidable.

Theorem (B-Sablik, 2016)

*If G is finitely generated, has decidable word problem and $d > 1$.
Then $G \rtimes_{\varphi} \mathbb{Z}^d$ admits a SA SFT.*

Ex: The discrete Heisenberg group.

Strongly aperiodic SFTs!

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Theorem (B, 2017)

If G_i are at least three infinite and finitely generated groups with decidable word problem. Then $G_1 \times \cdots \times G_n$ admits a SA SFT.

Ex: (not an obvious corollary) The Grigorchuk group.

A negative result

Recall the subshift:

$$X_{\leq 1} = \{x \in \{0, 1\}^G \mid 0 \notin \{x_g, x_h\} \implies g = h\}$$

configurations $x \in X_{\leq 1}$ have at most one occurrence of 1.

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configurations $x \in X_{\leq 1}$ have at most one occurrence of 1.

Theorem (Aubrun-B-Sablik 2015)

If G is finitely generated and recursively presented, then $X_{\leq 1}$ is effectively closed if and only if G has decidable word problem.

In particular, there are groups for which $X_{\leq 1}$ is not even conjugate to the subaction of an effectively closed subshift.

Computability in group invariants of shift spaces

Motivation

Given a subshift (X, σ) , its automorphism group is given by

$$\text{Aut}(X) = \{\phi : X \rightarrow X \text{ homeomorphism}, [\sigma, \phi] = \text{id}\}$$

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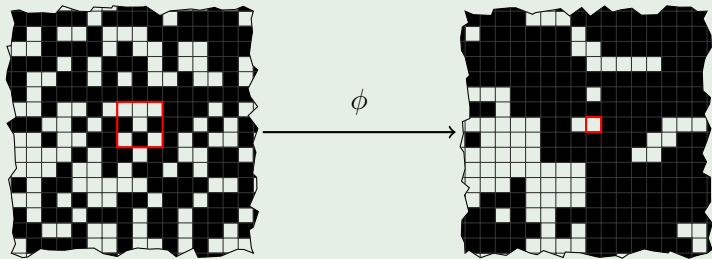
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Example and non-example.



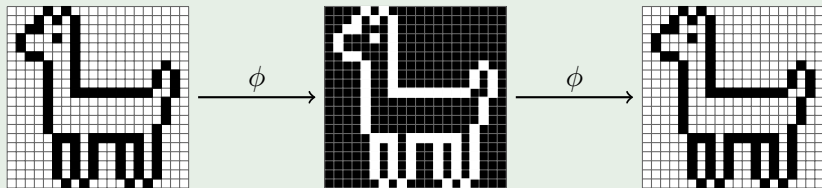
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The topological fullgroup of a dynamical system (X, T) where $T : G \curvearrowright X$ is

$$[[T]] = \{\phi \in \text{Homeo}(X) \mid \exists s : X \rightarrow G \text{ continuous, } \phi(x) = T^{s(x)}(x)\}.$$

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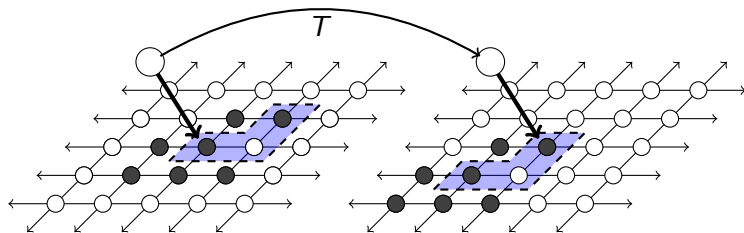
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This group is the restriction of a larger group introduced as an invariant of orbit equivalence.

In the case of a subshift it can be interpreted as a group of abstract Turing machines which are globally reversible and do not change the tape.

Motivation



$$s(\bullet \overset{\circ}{\circ}) = (1, 1)$$

It might be a good idea to understand these groups from a computability point of view:

Definition (three problems)

Let $G = \langle S \mid R \rangle$ be a finitely generated group.

- **Word problem:** Given $w \in S^*$, is w the identity of the group?
- **Torsion problem:** Given $w \in S^*$, is there $n \in \mathbb{N}$ such that w^n is the identity of the group?
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The real problem is that $[[\sigma]]$ and $\text{Aut}(X)$ in general are not finitely generated...

Two weird examples

▷ $K \subset \mathbb{N}$ be r.e. but undecidable.

Example 1: decidability depends on the presentation.

$$\begin{aligned} G &\cong \langle \{a_n\}_{n \in \mathbb{N}} \mid \{[a_n, a_m]\}_{n, m \in \mathbb{N}} \cup \{(a_k)^2\}_{k \in K} \rangle \\ &\cong \langle \{b_n\}_{n \in \mathbb{N}} \mid \{[b_n, b_m]\}_{n, m \in \mathbb{N}} \cup \{(b_\ell)^2\}_{\ell \text{ is even}} \rangle. \end{aligned}$$

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Example 2: No presentation with decidable WP, all f.g. subgroups have decidable WP.

▷ $\{p_n\}_{n \in \mathbb{N}} = \text{primes } 1 \pmod{4}$.

▷ $\{q_n\}_{n \in \mathbb{N}} = \text{primes } 3 \pmod{4}$.

$$G = \langle \{a_n\}_{n \in \mathbb{N}} \mid \{[a_n, a_m], (a_n)^{p_n q_n}\}_{n, m \in \mathbb{N}} \cup \{(a_k)^{p_k}\}_{k \in K} \rangle.$$

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For any finite alphabet A with at least two symbols, $\text{Aut}(A^{\mathbb{Z}})$ contains a finitely generated subgroup with undecidable torsion problem.

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Theorem (B-Kari-Salo, 2016)

The same is true for the topological full group of $(A^{\mathbb{Z}^d}, \sigma)$ if and only if $d \geq 2$.

Short and middle term

- Simulation theorem $G \times \mathbb{Z}$ or $G \times H$ for expansive actions.
- Strongly aperiodic subshifts in products $G_1 \times G_2$? (with $\text{WP}(G_i)$ decidable)
- Minimal effectively closed strongly aperiodic subshifts?
- Other applications: entropies of SFTs, realize mixing conditions, etc.
- Study unavoidable sets of patterns.
- Dynamical proof of Highman's theorem.

Long term (related to this talk)

- Characterize the groups which admit strongly aperiodic SFTs.
- Characterize the entropies of SFTs on f.g. groups (amenable... or maybe sofic).

Long term (unrelated to this talk)

- The equal entropy SFT cover problem.
- Domino problem for f.g. groups
- Better understand conjugacy of automorphism groups: the $\text{Aut}(\{0, 1\}^{\mathbb{Z}})$ vs $\text{Aut}(\{0, 1, 2\}^{\mathbb{Z}})$ problem.

Thank you for your attention!

Merci pour votre attention !

¡Gracias por su atención!



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Example

If we write $x = x_0x_1x_2x_3\dots$ we obtain,

$$\Psi(x) = \dots \$x_0\$x_1x_0\$ \$x_0\$x_2x_0\$x_1x_0\$ \$x_0\$ \$x_0\$x_1x_0\$ \$x_0\$x_3x_0\dots$$

How does one prove such a thing?

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How does one prove such a thing?

- ▷ pick a finite set of generators S of G .
- ▷ construct a subshift Π where every configuration is an S -tuple of configurations of the previous form.

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Claim

If T is an effectively closed action, Π is effectively closed.

How does one prove such a thing?

- ▷ Take Π and construct a sofic $\mathbb{Z}^2$ subshift $\tilde{\Pi}$ having Π in every horizontal row using the expansive simulation theorem.

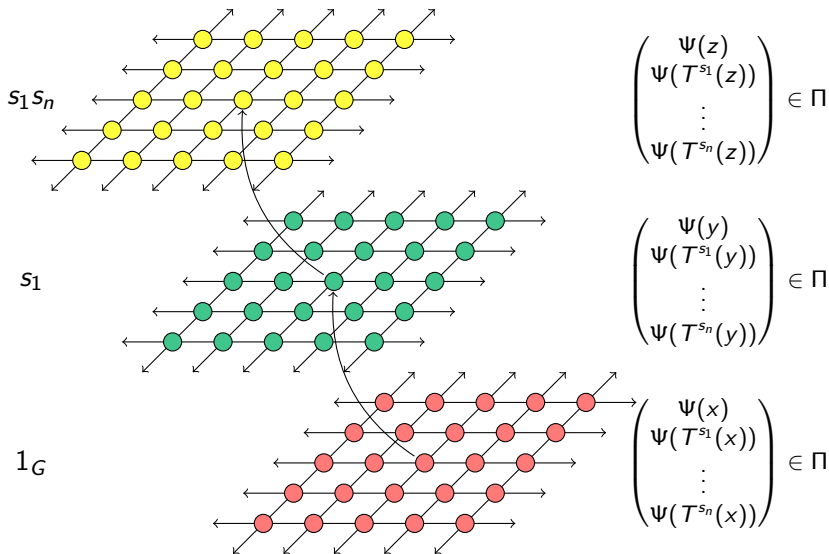
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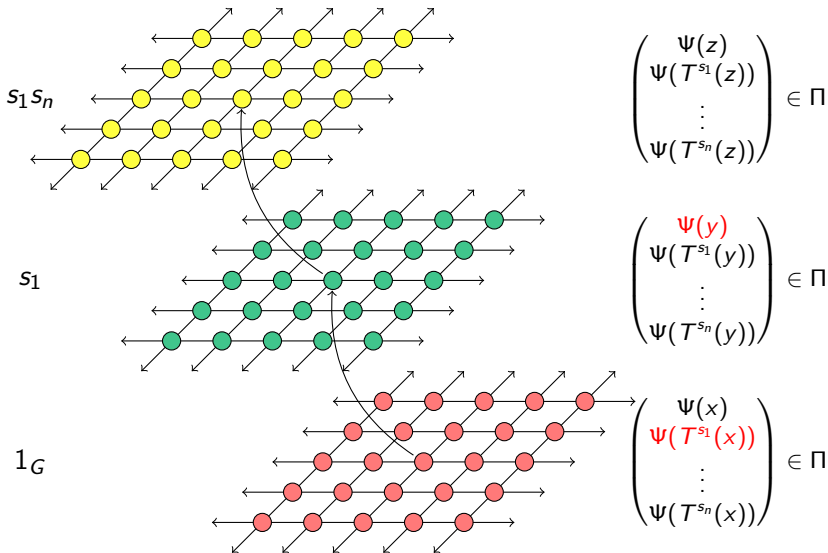
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- ▷ Put in every G -coset of $G \times \mathbb{Z}^2$ a configuration of $\tilde{\Pi}$.

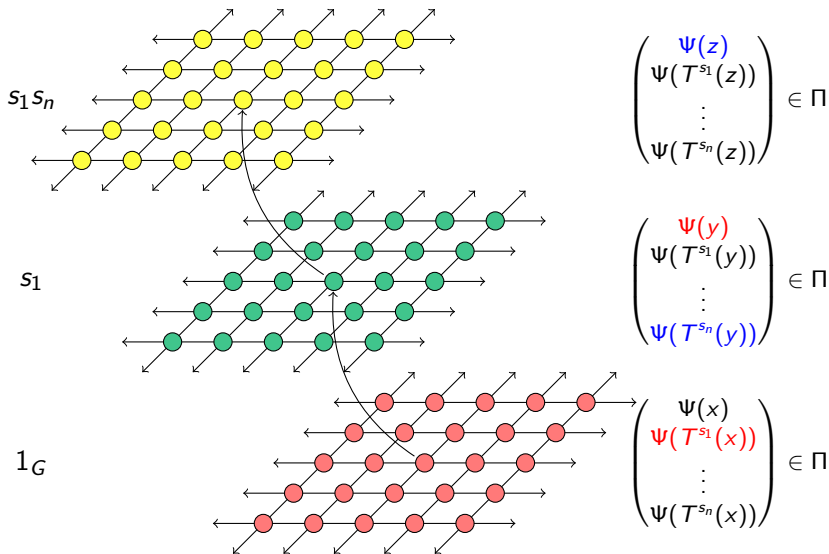
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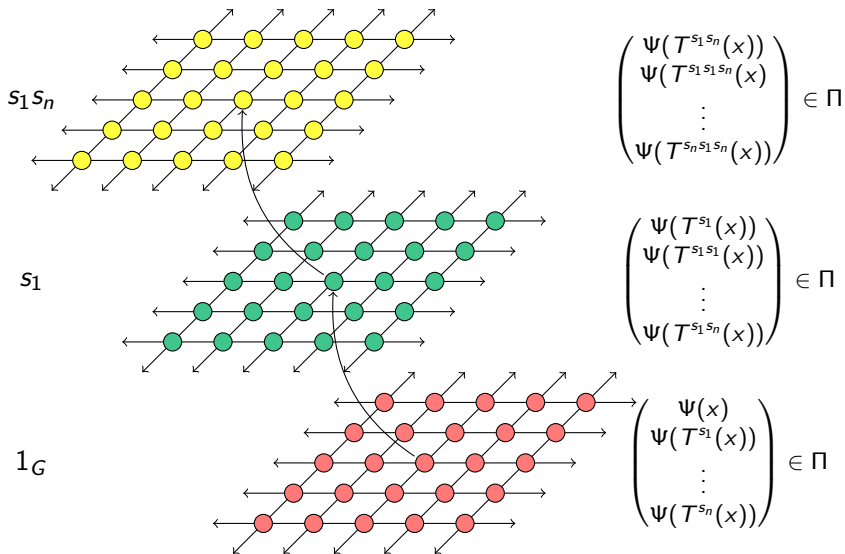
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Lovász Local Lemma (Asymmetrical version)

Let $\mathcal{A} := \{A_1, A_2, \dots, A_n\}$ be a finite collection of measurable sets in a probability space $(X, \mu, \mathcal{B})$. For $A \in \mathcal{A}$, let $\Gamma(A)$ be the smallest subset of $\mathcal{A}$ such that A is independent of the collection $\mathcal{A} \setminus (\{A\} \cup \Gamma(A))$. Suppose there exists a function $x : \mathcal{A} \rightarrow (0, 1)$ such that:

$$\forall A \in \mathcal{A} : \mu(A) \leq x(A) \prod_{B \in \Gamma(A)} (1 - x(B))$$

then the probability of avoiding all events in $\mathcal{A}$ is positive, in particular:

$$\mu \left(X \setminus \bigcup_{i=1}^n A_i \right) \geq \prod_{A \in \mathcal{A}} (1 - x(A)) > 0.$$

A sufficient condition for being non-empty

Let G a countable group and $X \subset \mathcal{A}^G$ a subshift defined by the set of forbidden patterns $\mathcal{F} = \bigcup_{n \geq 1} \mathcal{F}_n$, where $\mathcal{F}_n \subset \mathcal{A}^{S_n}$. Suppose that there exists a function $x : \mathbb{N} \times G \rightarrow (0, 1)$ such that:

$$\forall n \in \mathbb{N}, g \in G, \mu(A_{n,g}) \leq x(n, g) \prod_{\substack{gS_n \cap hS_k \neq \emptyset \\ (k,h) \neq (n,g)}} (1 - x(k, h)),$$

where $A_{n,g} = \{x \in \mathcal{A}^G : x|_{gS_n} \in \mathcal{F}_n\}$ and μ is any Bernoulli probability measure on $\mathcal{A}^G$. Then the subshift X is non-empty.

We say $x \in \{0,1\}^G$ has *the distinct neighborhood property* if for every $h \in G \setminus \{1_G\}$ there exists a finite subset $T \subset G$ such that:

$$\forall g \in G : x|_{ghT} \neq x|_{gT}.$$

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Proposition

If x has the distinct neighborhood property then $\overline{\text{Orb}_\sigma(x)}$ is strongly aperiodic.

Proof of the theorem

It suffices to show that there is $x \in \{0, 1\}^G$ with the distinct neighborhood property.

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Ingredients

- ▶ A constant $C \in \mathbb{N}$.
- ▶ An enumeration $s_1, s_2, \dots$ of G .
- ▶ $(T_i)_{i \in \mathbb{N}}$ a sequence of finite subsets of G such that for every $i \in \mathbb{N}$, $T_i \cap s_i T_i = \emptyset$ and $|T_i| = C \cdot i$.
- ▶ The uniform Bernoulli measure μ
- ▶ $\mathcal{A} := \{A_{n,g}\}_{n \geq 1, g \in G}$
- ▶ $A_{n,g} = \{x \in \{0, 1\}^G \mid x|_{gT_n} = x|_{gs_n T_n}\}$
- ▶ $x(A_{n,g}) := 2^{-\frac{Cn}{2}}$

Proof: On the blackboard.